will help to understand the energy conversions going along with the change from plasma to the thermodynamic states of supercritical water and hot vapor in the expanding bubble.

Sub-micrometer spatial resolution of shock wave emission and early bubble expansion requires novel techniques for laser-based speckle-free stroboscopic photography with picosecond exposure times, which are presently under development in our group. Highly reproducible breakdown events can be recorded by stroboscopic photography [40], whereas multi-exposure single-frame photography enables to capture also events with lower reproducibility. This approach was pioneered by Alcock et al. [376] (with high-quality prints in [377]) and has recently been sophisticated using illumination pulse bursts from a fiber laser with time separations down to 5 ns [378,379]. Availability of bursts consisting of about 10 pulses with ≈ 1 ns time separation will enable precise single-shot recordings of the early shock wave formation and emission. As described in section 5.8, the pressure evolution $p_s(r)$ can be derived from measured $r_s(t)$ data, and the shock wave energy can be obtained by integrating over the energy dissipation, which is encoded in the $p_S(r)$ curves [114].

Confocal probe beam scattering is a convenient method for single-shot measurements of cavitation bubble dynamics with high temporal resolution. While transmission measurements merely yield the bubble oscillation time $T_{osc,}$ we have demonstrated that also radius time curves can be acquired by recording the back reflected scattering signal. Interference of reflections from the front and rear bubble walls yields a signal from which $R(t)$ can be deduced [242].

Availability of $R(t)$ and $p_s(r)$ data from single shot recordings in combination with spectroscopic information on the evolution of plasma temperature and electron density will provide a data base that can advance the development of numerical models on plasma formation and early hydrodynamic events. Single-shot methods for $R(t)$ and $r_s(t)$ also open new avenues for the investigation of optical breakdown and cavitation events in viscoelastic materials in general and transparent biological tissues and cells in particular [380,381]. They are of great interest for cavitation rheology that to date relies on stroboscopic photography [382,383] or high-speed videography [384,385].

For a visualization of processes inside the cavitation bubble at a solid target during LAL and LIBS, a good view into the bubble with high spatiotemporal resolution is of major importance. For optical imaging, this can be achieved by Koehler illumination of the breakdown region at large numerical aperture [105,386]. Phase contrast X-ray imaging also offers a view through bubbles [91,387,388], while time-resolved small-angle X-ray scattering provides additionally the particle distribution inside the bubble [74,91,389].

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6.2. Roadmap for multiscale modelling of plasma formation and hydrodynamic events

The conceptual framework and experimental results provided in this paper motivate to establish a unified model of plasma formation, shock wave emission, and bubble generation in water. Such comprehensive model should portray the interplay of MPI, AI, recombination and TI, and link it to the modeling of hydrodynamic events in a way that the experimental results of Figs. 4-6 with its different breakdown scenarios are reproduced. Combining a breakdown model with simulations of spherical bubble generation in a compressible liquid [42,390] is a promising approach for achieving this goal because the relative simplicity of such models facilitates parametric studies in (τ_L, λ_L, E_L) space.

On a higher level of sophistication, modeling should consider the spatiotemporal evolution of the energy density distribution in the breakdown region as starting point for the hydrodynamic part of the modeling chain including shock wave emission and bubble oscillations. Advanced models of femtosecond optical breakdown in bulk media covering the interplay of beam nonlinear propagation and nonlinear absorption are already available [130,131,136,140,141,289]. However, modeling approaches covering the features of strings, hot spots, and radiative energy transport relevant for nanosecond breakdown (Figs. 7, 8) are still lacking.

Remarkably, our heuristic considerations based on generic equations assuming a constant band structure of a transparent dielectric could delineate different breakdown scenarios in water that were experimentally observed in a large range of pulse energies and plasma energy densities. However, future modeling must consider that this approach is fully appropriate only in the low-energy density regime close to the optical breakdown threshold and becomes insufficient when the target material disintegrates and an electron-ion plasma forms.

For cavitation produced by the expansion of an electron-ion plasma, a dual EOS approach assuming a gaseous domain inside the plasma/bubble volume and a liquid EOS outside, which is used in most models, does not cover the real complexity of events. Recently, the use of a Noble-Abel stiffened gas EOS for bubble interior and exterior has become popular, in which a covolume is introduced that allows to account for repulsive short-distance effects linked to molecular motion [391]. This EOS resolves the temperature inaccuracy associated with the commonly employed Tait EOS for liquids [392], and allows to express also a real-gas EOS by appropriate parameter choice [43]. However, it does not overcome the dual EOS approach itself. Particle-based approaches and incorporation of chemical reactions [296,393] promise more flexibility and accuracy for modeling the plasma-vapor transition - at the cost of increased complexity. While chemical processes driven by high temperature and pressure in electron-ion plasmas

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have been considered in the modeling of bubble collapse in SBSL and sonochemistry [183,394-396], their integration in the modeling of laser-induced plasma formation and bubble generation is still in its infancy [393].

Information on the spatiotemporal dynamics of shock wave emission and energy partitioning arising from an inhomogeneous energy distribution in the plasma can be gained through ***finite volume modeling*** of the hydrodynamic events [64,67,397-401]. They are also a versatile tool for modeling laser-induced cavitation under asymmetric boundary conditions at solid targets such as in LAL and LIBS. At present, they often assume isotropic energy distribution into a spherical start volume and neglect heat diffusion as well as phase transitions at the bubble wall but the integration of spatiotemporal features of energy deposition and heat and mass transfer is a field of active research. Information about the spatiotemporal evolution of the laser-induced energy density distribution that define the starting conditions for refined finite volume modeling can be obtained either from high-resolution plasma and stroboscopic photographs, or through numerical simulations of plasma formation. Calculations of the deposited energy distribution have been performed for fs breakdown [136,215,252] but still pose a challenge for ns breakdown, because neither the spatiotemporal instabilities of avalanche ionization leading to string formation nor radiative energy transport are yet included in current models.

Dehghani et al. [402] and Preso et al. [403] recently reviewed perspectives and challenges in 'green' applications of sonochemistry. Sonochemical reactions rely on the high pressure and temperature developed during bubble collapse [74,394,396]. Like single bubble sonoluminescence [183], also laser-induced cavitation is a model system for detailed investigations of bubble collapse under controlled conditions. Collapse behavior is tunable through variation of breakdown plasma features as they govern the chemical reactions taking place upon breakdown, which determine the amount and composition of permanent gas during the collapse phase. A large gas content cushions the collapse and reduces the collapse pressure, while pure vapor bubbles collapse more vigorously. It was recently shown that a large energy density upon breakdown goes along with lower collapse pressure than a moderate energy density – likely due to an increased rate of molecular dissociation and permanent gas production in hotter plasmas [42,66].

Generally, understanding the physicochemical interactions connected with laser-induced breakdown and cavitation is essential for many applications and deserves more attention in the Physics community. Tracking the chemical reaction chain from optical breakdown to collapse with its connection to

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